

SMARTBRIDGE

Let's Bridge the Gap | SKILL WALLET - A SmartBridge Product

Date	29-09-2026
Team ID	SWTID-2026-7949
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value

Problem-Solution Fit Canvas

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] • Academic Faculty & Branch Coordinators (EEE Department). • ServiceNow System & Security Administrators. • Academic Operations Staff & Department Data Entry Operators.	6. CUSTOMER LIMITATIONS [CL] • Inability to modify global ACL scripts without elevated security_admin privileges. • Restricted technical scripting knowledge among departmental end users. • Dependency on centralized IT for record-level permission provisioning.	5. AVAILABLE SOLUTIONS PROS & CONS [AS] • UI Policies / Client Scripts: Pros: Quick setup. Cons: Insecure; easily bypassed via REST API or list export. • Before-Query Business Rules: Pros: Fast row filtering. Cons: Does not enforce fine-grained CRUD (Create, Write, Delete) security.
2. PROBLEMS/PAINS + FREQUENCY [PR] • Unrestricted cross-department student data exposure (ECE, CSE visible to EEE). • Risk of accidental deletion/editing of records by non-owning branches. • Frequency: Constant (every query, page load, and search transaction).	9. PROBLEM ROOT / CAUSE [RC] • Lack of server-side Script-Controlled Access Control Lists (ACLs) evaluated against row-level field values (Branch == 'EEE'). • Over-permissive baseline read/write access on custom tables (u_institution_details).	7. BEHAVIOR + INTENSITY [BE] • Faculty frequently attempt to filter lists manually using UI search bars. • High intensity frustration and anxiety regarding student PII data leaks and compliance breaches.
3. TRIGGERS TO ACT [TR] • Institutional security audit identifying unauthorized cross-department data access. • Deployment of the new Institution Details custom table requiring zero-trust row segregation.	10. YOUR SOLUTION [SL] • Deploy Script-Controlled Read ACL evaluating branch condition (Branch == 'EEE') & role (gs.hasRole('bb1')). • Configure granular CRUD ACLs: bb2 (Create), bb3 (Write), bb4 (Delete), with full admin override.	8. CHANNELS OF BEHAVIOR [CH] • Online: ServiceNow Next Experience Portal, Application Navigator lists, REST APIs. • Offline: Departmental IT audits, governance reviews, faculty meetings.
4. EMOTIONS BEFORE/AFTER [EM] • Before: Anxious about privacy leaks, confused by cross-department clutter, distrustful of system security. • After: Confident, secure, reassured by zero-trust boundaries, productive with clean departmental views.	10. SOLUTION IMPACT & VERIFICATION • Server-side enforcement guarantees 0% record leakage across UI, APIs, and exports. • Strict least-privilege CRUD validation confirmed via user impersonation testing for admin and EEE users. • Seamless scalability for additional departments (e.g., MECH, CIVIL).	KEY VALUE DELIVERED • Complete data privacy compliance (FERPA & organizational policy). • Automated, server-side access governance without manual filtering. • Fully auditable security architecture via ServiceNow Studio.